

DIABETES RISK ANALYSIS

Pima Indians Diabetes Dataset

A Health Data Analytics Report

Prepared by: Alabi Adeola Dominion

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Tools used: Python, Pandas, Numpy, Matplotlib, Seaborn

Dataset Source: UCI Machine Learning Repository

1. INTRODUCTION

This report presents a health data analysis of the Pima Indians Diabetes Dataset obtained from the UCI Machine Learning Repository. The dataset contains medical records of 768 female patients of Pima Indian heritage, including clinical measurements such as glucose levels, BMI, blood pressure, insulin, and age. The primary objective of this analysis is to identify key risk factors associated with

diabetes and provide actionable insights for clinical decision making.

2. DATASET DESCRIPTION

The dataset consists of 768 patients and 9 variables. All patients are female, at least 21 years old, and of Pima Indian heritage. The variables included in the dataset are:

- Pregnancies — Number of times pregnant
- Glucose — Plasma glucose concentration
- Blood Pressure — Diastolic blood pressure (mm Hg)
- Skin Thickness — Triceps skin fold thickness (mm)
- Insulin — 2-hour serum insulin (mu U/ml)
- BMI — Body mass index (weight in kg/height in m²)
- Diabetes Pedigree — Diabetes family history score
- Age — Age in years
- Outcome — Diabetes diagnosis (0 = No, 1 = Yes)

Initial exploration revealed data quality issues including impossible zero values in clinical columns such as Glucose, Blood Pressure, BMI, Skin Thickness, and Insulin. These were identified and treated during the data cleaning.

3. DATA CLEANING

Prior to analysis, the dataset was examined for data quality issues. The following cleaning steps were performed:

- **Missing Values:** Five clinical columns contained impossible zero values — Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI. These zeros were replaced with NaN (Not a Number) to properly represent missing data.
- **Mean Imputation:** Missing values were replaced with the column mean to preserve dataset size while maintaining statistical integrity.
- **Duplicate Records:** The dataset was checked for duplicate patient records. No duplicates were found.

After cleaning, all 768 patient records were retained with no missing values across any column. The dataset was confirmed clean and ready for analysis.

4. ANALYSIS AND FINDINGS

4.1 Diabetes Prevalence

Analysis revealed that 268 out of 768 patients (34.90%) were diagnosed with diabetes. This prevalence is significantly higher than the global diabetes average of approximately 10.5%, reflecting the well-documented high diabetes risk among the Pima Indian population.

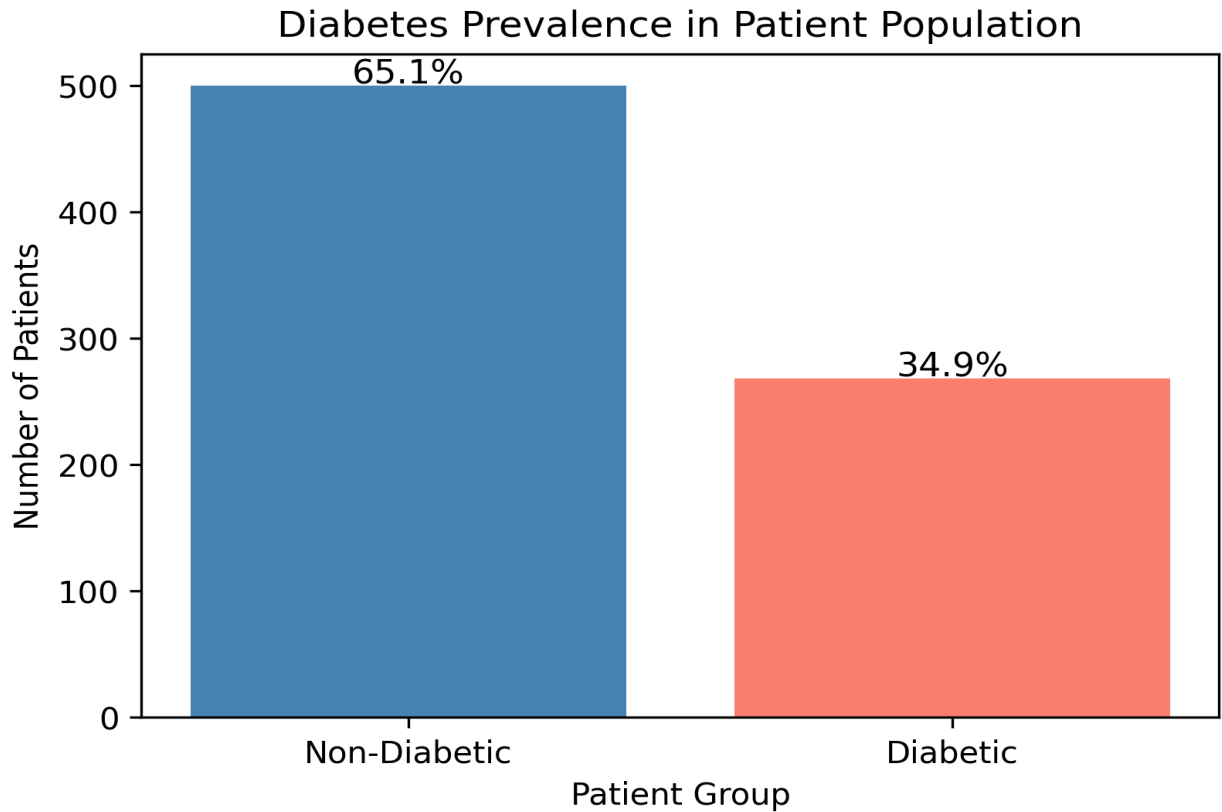


Figure 1: Diabetes Prevalence among 768 patients

4.2 Comparison Of Diabetic and Non-Diabetic Patients

A comparison of key clinical indicators between diabetic and non-diabetic patients revealed consistent differences across all measured variables:

- **Glucose:** Diabetic patients showed 28% higher average glucose levels (142.17) compared to non-diabetic patients (110.71) — the largest difference observed across all indicators.
- **BMI:** Diabetic patients had a higher average BMI (35.38) compared to non-diabetic patients (30.89), confirming the strong link between obesity and diabetes.
- **Age:** Diabetic patients were on average 6 years older (37.07 years) than non-diabetic patients (31.19 years), confirming age as a significant diabetes risk factor

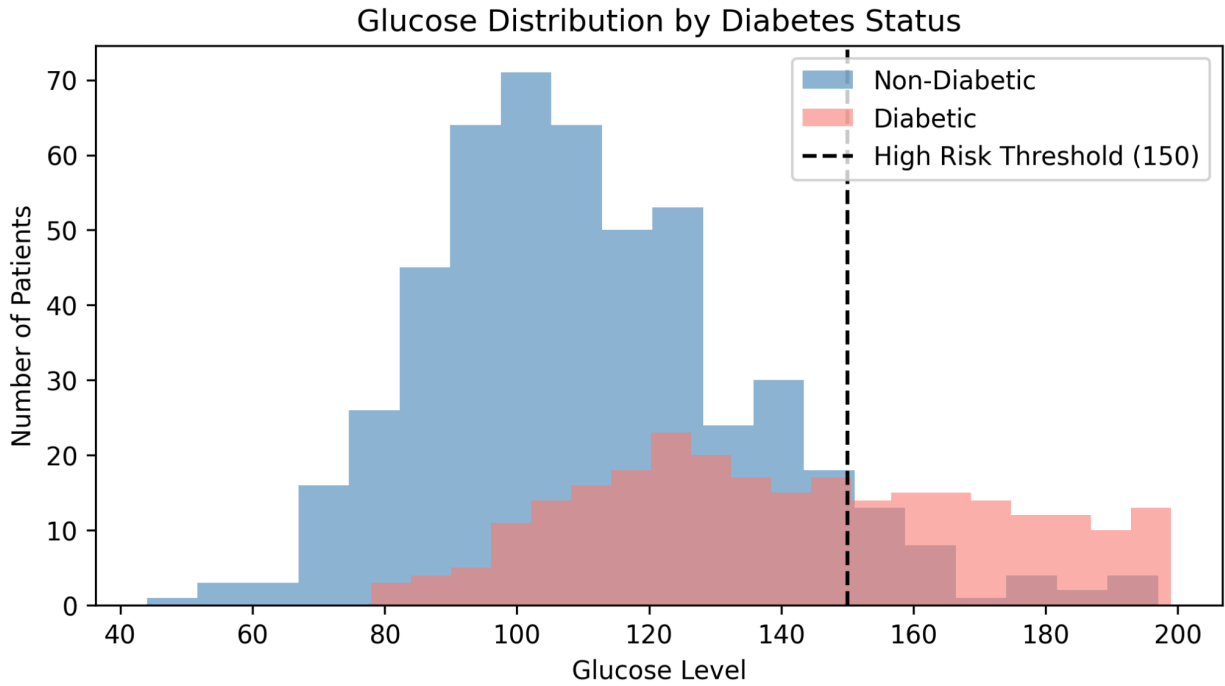


Figure 2: Glucose Distribution by Diabetes Status

4.3 Correlation Analysis

Correlation analysis identified the following factors as the strongest predictors of diabetes outcome:

- Glucose — 0.49 (strongest predictor)
- BMI — 0.31 (second strongest)
- Age — 0.24 (third strongest)
- Pregnancies — 0.22
- Blood Pressure — 0.17 (weakest predictor)

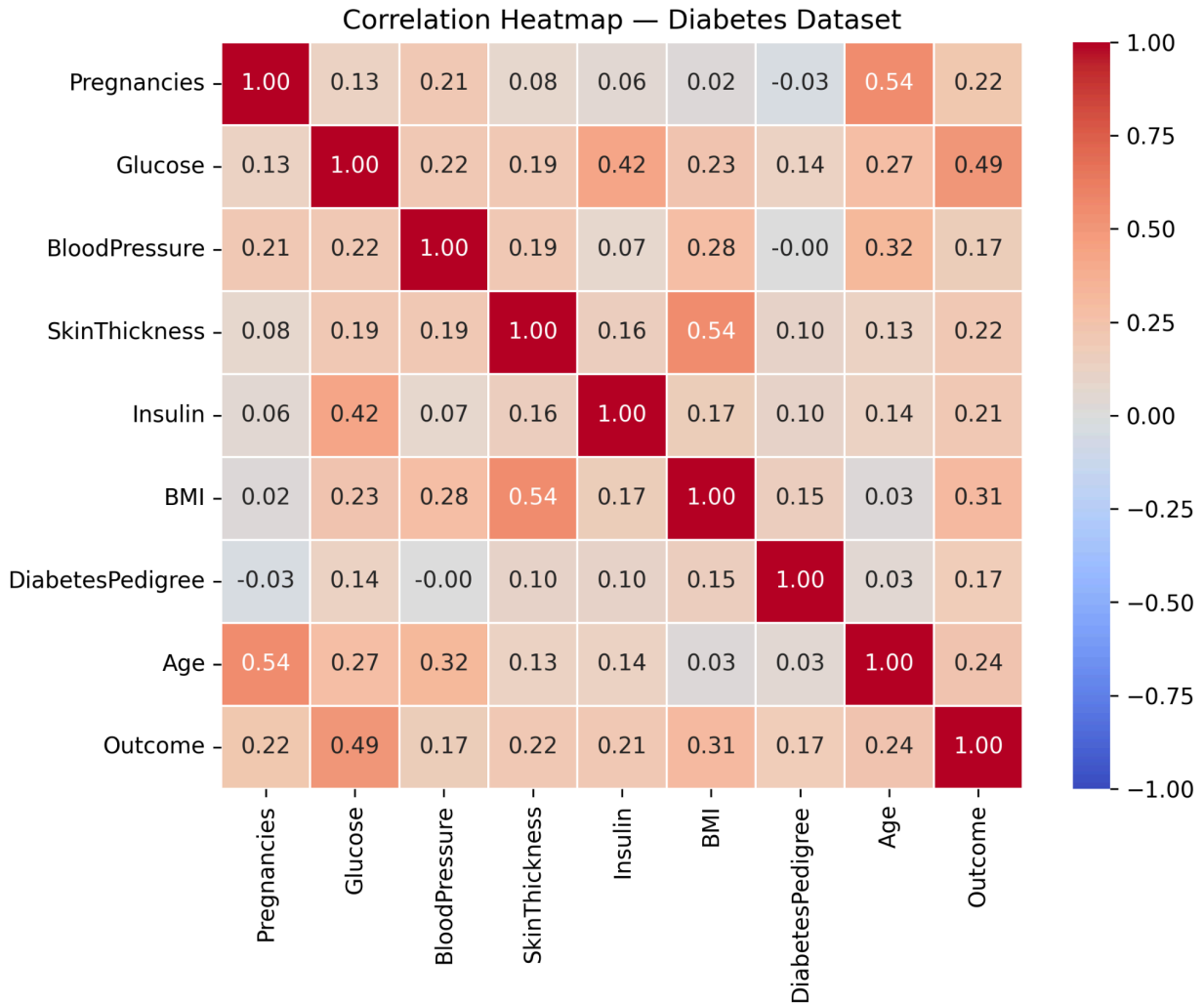


Figure 3: Correlation Heatmap of Clinical Indicators

These findings confirm that glucose level and BMI are the most reliable indicators of diabetes risk in this patient population.

5. CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This analysis of 768 Pima Indian female patients reveals that diabetes prevalence in this population (34.90%) is significantly above the global average. Glucose level and BMI emerged as the strongest predictors of diabetes risk, with diabetic patients consistently showing higher values across all measured clinical indicators. Age was also identified as a significant contributing factor, with diabetic patients being on average 6 years older than non-diabetic patients.

5.2 Recommendations

Based on the findings of this analysis, the following recommendations are proposed:

- **Glucose Monitoring:** Patients presenting with glucose levels above 150 mg/dL should be prioritized for immediate diabetes screening and management.
- **Weight Management Programs:** Given the strong link between BMI and diabetes, targeted weight management and nutritional intervention programs should be implemented for patients with BMI above 30.
- **Age-Based Screening:** Patients aged 35 and above should be flagged for routine diabetes screening given the observed relationship between age and diabetes risk.
- **Further Research:** Future analysis should investigate the combined effect of multiple risk factors simultaneously to develop a more precise diabetes risk scoring model for this population.

6. DATA SOURCES

Dataset: Pima Indians Diabetes Database

Source: UCI Machine Learning Repository

Link: kaggle.com/datasets/uciml/pima-indians-diabetes-database

Tools: Python 3, Pandas, NumPy, Matplotlib, Seaborn

Analysis Environment: Google Colaboratory